METHODOLOGY WHITE PAPER

GEMS 2026

Gaming Economic Modeling System

The Casino Economic Impact Model

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June 2026

Abstract

GEMS (Gaming Economic Modeling System) is a state-level economic impact model for casino and gaming operations covering all 50 US states. District of Columbia coefficients are not included in the deployed dataset. The model applies Input-Output (IO) analysis under the Industry Technology Assumption, using EPA State Input-Output tables, BLS Quarterly Census of Employment and Wages data, and gambling-specific coefficients derived from BEA Detail IO tables. For a given revenue profile, the model estimates direct, indirect, and induced effects on economic output, GDP (value added), filled jobs, estimated FTEs, employee compensation, and tax revenue. This white paper documents the modeling framework, data sources, and key methodological decisions of the 2026 edition, and describes the model's appropriate uses and limitations.

Suggested citation

Philander, K. (2026). GEMS: Gaming Economic Modeling System (Version 2026) [Computer software]. GP Consulting. https://gamblingpolicy.com/tools/economic-impact/

When citing results produced by the model, please also state the analysis date and the inputs used (state, property type, and revenue assumptions), as results depend on user-supplied inputs.

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1. Introduction

Casino developments are routinely evaluated on their wider economic contribution: the jobs they support, the value they add to regional GDP, and the tax revenue they generate. Yet general-purpose economic multipliers are poorly suited to gaming. Published entertainment and recreation multipliers blend casinos with golf courses, fitness centers, and amusement parks - businesses with fundamentally different cost structures and labor intensities.

GEMS addresses this gap. It combines state-specific Input-Output multipliers with gambling-specific industry coefficients and department-specific revenue structures to produce defensible, state-level estimates of the economic footprint of casino and gaming operations. The model supports land-based property types (casino hotels, stand-alone casinos, slot parlors, and bar/restaurant gaming) as well as online casino and sports betting operations.

For each analysis, the model reports six metrics - gross output, GDP (value added), filled jobs, estimated FTEs, employee compensation, and tax revenue - each decomposed into direct, indirect, and induced components with the implied multipliers.

2. Modeling Framework

2.1 Industry Technology Assumption

The model is built on the Industry Technology Assumption (ITA) framework of Input-Output economics. The direct requirements matrix is constructed from the Make and Use tables:

$$A = D \times B$$

where **D** is the market share matrix derived from the Make table (the share of each commodity produced by each industry) and **B** is the commodity coefficient matrix derived from the Use table (the amount of each commodity required per dollar of output). The result **A** is an industry-by-industry direct requirements matrix.

2.2 Leontief Inverse and Type I Multipliers

$$L = (I - A)^{-1}$$

The Leontief inverse captures all rounds of inter-industry purchasing. Each column sum gives the Type I output multiplier - total output generated per dollar of direct output, including all supplier (indirect) effects.

2.3 Type II Augmentation

The A matrix is augmented with a household row (labor income coefficients) and household column (consumer spending patterns). The augmented Leontief inverse yields Type II multipliers, which additionally capture induced effects from the spending of labor income earned in direct and indirect activity.

3. Effect Decomposition

Effect	Source	Example
Direct	The initial economic activity	Casino hires employees, pays compensation, generates revenue
Indirect	Supply-chain purchasing	Casino buys food, linens, and equipment from suppliers; suppliers buy from their suppliers
Induced	Household spending	Casino and supplier employees spend labor income on housing, groceries, and services

Effects are decomposed from the multiplier pairs as follows:

```
Direct output = revenue
Indirect output = revenue x (Type I - 1)
Induced output = revenue x (Type II - Type I)
Total output = revenue x Type II
```

The same decomposition applies to GDP, employee compensation, and TOPI using their respective multiplier pairs.

4. Employment Estimation

Employment is estimated with a consistent-2023 employment satellite rather than a simple output multiplier, because a dollar of indirect or induced GDP generates more or fewer jobs depending on whether it flows to labor-intensive sectors (restaurants) or capital-intensive sectors (utilities):

```
Direct jobs = Direct output x Jobs_per_Output
Supplier or induced jobs = GDP_effect x
Effect_Specific_Jobs_per_GDP
```

Coefficient	Applies to	Reflects
Direct_Jobs_Per_1M_Output	Direct employment	2023 jobs per $1M of 2023 output
Emp_Coef	Direct diagnostic	2023 jobs per $1M of 2023 value added
Indirect_Emp_Coef	Indirect employment	Weighted average labor intensity of supplier industries
Induced_Emp_Coef	Induced employment	Weighted average labor intensity of household-spending industries

QCEW jobs are divided by 2023 StateIO output and value added. Jobs/output is propagated through the Leontief output-requirements matrix; supplier and induced jobs/GDP coefficients are derived from the resulting job totals and corresponding GDP increments. QCEW annual average employment counts filled full-time and part-time jobs, including temporary and permanent positions; it does not use hours or convert positions to full-time equivalents. A person with multiple jobs may be counted more than once.
User-provided jobs replace only direct employment. User-provided cash wages replace the direct labor value and rebase the household-spending effects. The satellite is built from the official BLS annual singlefile with exact aggregate mappings. StateIO's synthetic HS, GFE, and GSLE sectors have no QCEW-compatible measure and are explicit zeros rather than average-sector imputations.

QCEW does not supply FTEs, so filled jobs remain the primary employment measure. The model reports a separate estimate using 2023 BEA NIPA tables 6.4D and 6.5D:

```
Estimated FTEs = Modeled jobs x (BEA FTE employees / BEA employee jobs)
```

The 71-sector national ratio crosswalk is versioned in fte_ratios_2023.csv. A separate FTE/output satellite is propagated through the Type I and Type II matrices, so direct, supplier, and induced FTEs use the ratios of the industries where the corresponding jobs occur. These are national industry estimates, not establishment-reported hours. Known job inputs are converted with the selected direct industry's ratio.

5. Reporting Basis & Inflation Adjustment

All monetary results the model reports - output, GDP, employee compensation, and tax revenue - are in current-year (nominal) dollars, the same dollars the user enters. The dollar multipliers and coefficients are unitless ratios applied directly to current-dollar revenue, so their results are automatically in current dollars. No reported dollar figure is deflated.

Employment is the one quantity that needs a base-year adjustment, because the employment coefficients carry a dollar denomination. They are calibrated with **2023** QCEW jobs and 2023 StateIO output/value added. The model therefore restates the relevant output or GDP effect in 2023 dollars purely as an internal step in the jobs calculation:

```
Deflator = CPI_2023 / CPI_current_year
Jobs = (Output_or_GDP_current x Deflator) x Employment_Coefficient
```

This conversion affects the employment estimate only; it does not change any dollar figure the model displays. The 2019 StateIO year remains the production-structure vintage, not the employment coefficient price year.

The CPI table uses official BLS calendar-year annual averages. When the analysis year is newer than the latest complete annual average, GEMS uses the latest complete year (currently 2025) rather than presenting a partial-year index as an annual value.

6. Gambling-Specific Adjustments

The calibration separates geography from industry composition. EPA StateIO supplies each state's supplier relationships, import leakages, and Leontief structure. National BEA detail data supplies ratios describing how gambling differs from its broader parent sector. The ratios adapt the state model rather than replacing it with a generic local hospitality multiplier.

Published IO tables contain only the blended sector 713 (Amusement, Gambling, and Recreation). The single canonical generator builds USEEIOv2.0.1-411 and compares detailed gambling (713200/US) with an output-weighted 713 parent:

Component	Gambling-to-parent ratio
Direct value added	1.008
Direct employee compensation	0.863
Direct TOPI	0.609
Direct jobs per output	0.540
Supplier output increment	0.967
Supplier value-added increment	0.993
Supplier compensation increment	1.047
Supplier TOPI increment	0.940

The generator applies direct ratios and supplier-increment ratios separately while retaining StateIO state variation. Because USEEIO is an open Type I model, the same documented supplier ratios scale StateIO induced increments. Supplier and induced jobs/GDP coefficients remain tied to the industries where those effects occur.

The former competing legacy calibration has been removed. All land-based property labels use this canonical gambling series; department inputs introduce lodging, food, and other composition.

7. Tax Revenue Estimation

The model computes three categories of tax revenue:

7.1 Taxes on Production and Imports (TOPI)

Derived from the IO tables, TOPI flows through the Leontief inverse like employee compensation and captures sales and gross-receipts taxes, licenses, property taxes, excise taxes, and other production-related taxes across all direct, indirect, and induced activity.

7.2 Gaming Tax

A state-specific tax on Gross Gaming Revenue (GGR), sourced from state gaming commission reports and the AGA State of the States. Gross-receipts, amusement, and license taxes are already within the TOPI accounting concept. GEMS therefore reports statutory gaming tax as a memo component of direct TOPI and does not add it again to the non-duplicative tax total. GEMS does not subtract an estimated embedded amount because StateIO does not identify the state-specific gaming-tax portion of sector 713 TOPI. Supported structures:

A legacy land-based total-revenue input is not silently treated as GGR for this memo estimate. An explicit gaming-revenue input is required; online total revenue remains GGR by construction.

- **Flat rate** - a single percentage (e.g., Nevada at 6.75%)
- **Graduated tiers** - increasing rates at revenue thresholds (e.g., Colorado, Iowa)
- **Split by game type** - different rates for slots vs. table games (e.g., Pennsylvania)
- **Split-tiered** - separate graduated schedules for slots and tables (e.g., Illinois)

7.3 Payroll and Household Taxes

Employer-side payroll taxes (FICA, FUTA, SUTA, and state-specific SDI/PFML contributions) and household taxes are applied to estimated cash wages, not to the full V001 employee-compensation amount. Until state-industry supplements data are added, cash wages are estimated with the 2019 national BEA NIPA wage-and-salary share of employee compensation (81.44%). Household taxes use BEA personal current tax ratios (federal, state, and local).

8. Data Sources

Source	Vintage	Use
EPA State Input-Output (StateIO) models, via the stateior R package	2019 / 2023	2019 production structure; 2023 output/VA denominators for employment
BLS Quarterly Census of Employment and Wages (QCEW)	2023 annual averages	Filled-job counts used in employment coefficients
BEA NIPA Tables 6.4D and 6.5D	2023	National industry FTE/job ratios for estimated-FTE coefficients
BEA Detail IO tables via useeior (USEEIOv2.0.1-411)	2017	National gaming/property-to-parent ratios; source tables versioned in repository
State gaming commissions, AGA State of the States	Current	Gaming tax schedules
BLS CPI-U annual averages	Current	Inflation adjustment

Multipliers vary significantly across states due to differences in economic structure, supply-chain density, and trade patterns. States with more diversified, self-sufficient economies tend to have higher multipliers

because more supplier spending stays in-state.

9. Sector and Property-Type Mapping

Revenue entered by department is mapped to the corresponding IO sector with state-specific multipliers:

IO sector	NAICS	Casino department
711AS	711, 712	Other (entertainment, retail)
713	713	Gaming (GGR)
721	721	Lodging
722	722	Food & Beverage

All land-based property labels use the canonical gambling production function. When total revenue is entered, the complete amount uses that function and is an approximation if non-gaming revenue is material. Department mode separately assigns lodging, food, and other revenue to their corresponding sectors:

NAICS	Property type	Typical operations
721120	Casino Hotel	Full-service resort with gaming, lodging, F&B;
713210	Stand-alone Casino	Gaming facility without hotel
713290	Slot Parlor	Slot machines / VLTs only
722410	Bar/Restaurant Gaming	Gaming secondary to food/beverage service
-	Online Casino / Sports Betting	Remote gaming operations (separate multiplier sets)

The four land-based NAICS codes are reporting labels, not claims of four separately estimated production functions.

10. Limitations and Appropriate Use

- Results are **estimates** based on average inter-industry relationships; they should be interpreted as indicative rather than definitive.
- State-level analysis captures only impacts within the state boundary; spending that 'leaks' to other states is excluded. This yields conservative estimates of local benefits.
- IO models assume fixed production technology and no supply constraints; very large projects may alter local economic structure in ways the model does not capture.
- The model deliberately retains the 2019 pre-pandemic interindustry structure while using a consistent-2023 employment satellite. This is a structural hybrid, not a mixed labor numerator/denominator.

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- National gambling ratios, employment coefficients, the property assembler, deployed JSON, and offline integrity tests are versioned. The external QCEW input is the official BLS annual singlefile.
- Online-gambling adjustments are approximations based on a small public-company evidence base and retain land-based gambling supply-chain relationships.
- The model is appropriate for policy analysis, planning, and comparative assessment. It is **not** appropriate for precise forecasting, investment decisions without professional advice, or legal proceedings.

11. Versioning

The model version is the release edition year (e.g., GEMS 2026). When a new edition is published - whether from regenerated datasets or methodology updates - the version is bumped and this document is updated. Reports generated by the calculator embed the model version and the data vintages used, so results remain traceable.

Version	Released	Notes
GEMS 2026	2026	Initial release. 2019 EPA StateIO multipliers, 2023 QCEW employment, USEEIOv2.0.1-411 gambling adjustments.

12. References

American Gaming Association. *State of the States*. Annual editions.

Ingwersen, W., et al. *stateior* and *useeior* R packages. US Environmental Protection Agency. https://github.com/USEPA/stateior

Leontief, W. (1936). Quantitative input and output relations in the economic systems of the United States. *The Review of Economics and Statistics*, 18(3), 105-125.

Miller, R. E., & Blair, P. D. (2009). *Input-Output Analysis: Foundations and Extensions* (2nd ed.). Cambridge University Press.

US Bureau of Economic Analysis. Detail Input-Output tables (USEEIOv2.0.1-411).

US Bureau of Economic Analysis. Compensation of employees glossary and NIPA Table 1.12.

US Bureau of Economic Analysis. NIPA Tables 6.4D and 6.5D, Full-Time and Part-Time Employees and Full-Time Equivalent Employees by Industry.

US Bureau of Economic Analysis. Taxes on production and imports definition and GDP-by-state methodology.

US Bureau of Labor Statistics. Quarterly Census of Employment and Wages, 2023 annual averages.

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